

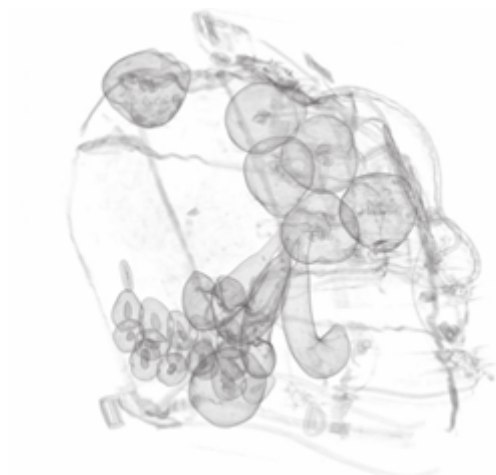
Through the Modern Seamless Border Clearance initiative, the department has established itself as the world leader in biosecurity innovation and in streamlining border clearance.

Australia is undertaking a ground-breaking trial using 3D x-ray technology in a biosecurity environment. The department was the first biosecurity agency in the world to deploy 3D x-ray capability in the airport environment. Under this project the department has also developed and deployed the world's first algorithm to automatically detect biosecurity risk material.

Following these early successes the department now has 3D x-ray units installed in Melbourne International Airport and Melbourne International Mail Centre. By mid-2020, 3D x-rays will also be operating at the Sydney International Mail Centre.

The innovative implementation of this technology has allowed the department to successfully increase detection of biosecurity risk items with 3D x-ray screening, resulting in double the detections in the airport and triple in the mail environment compared to current screening technologies.

What biosecurity risk can you see in the x-ray image below?



X-ray image of strawberries, bananas, apples and pears

Aim of the trial

Our aim is to develop an extensive 3-dimensional image library of biosecurity risk items. With these we have successfully developed the world's first algorithm for automatically detecting biosecurity risk material. These algorithms will allow the department to continue to modernise our border processes and explore opportunities for offshore/remote screening capabilities.

The 3D x-ray has reduced the number of 'false positives', reduced manual inspections, increased the rate of seizures and allows officers a higher level of confidence when clearing bags through x-ray.

When screened through a 3D x-ray, a smaller percentage of bags are opened for inspection, but a higher percentage of inspected bags contain actionable biosecurity risk material.

Even before the auto-detection algorithm was deployed, the 3D technology has resulted in more than twice as many seizures of meat products compared to existing 2D x-ray technology.

Real time tomography



Image of 3D X-ray unit, the Rapiscan Real Time Tomography RTT®110

The particular unit being trialed is the Rapiscan Real Time Tomography RTT®110, which has the ability to use the auto-detection algorithms for biosecurity risk screening.

Real time tomography uses 3-dimensional x-rays that produce images in real time as the item passes through the x-ray unit.

It differs from other x-ray technology as the image is taken without the need to rotate an x-ray emitter around the object.

Stakeholders

We are working alongside other Australian and New Zealand government border agencies and industry stakeholders to ensure the best outcomes of this innovation trial.

New Zealand is trialling the same Rapiscan 3D x-ray unit at Auckland Airport. The x-ray images generated from both the Australian and New Zealand trials will be combined to create an extensive biosecurity risk image library.

Further information

- [2018 Federal Budget Announcement – Modern Seamless Border Clearance](#)
- [Trans-Tasman Cooperation on Biosecurity Risk Detection Technology – Australia](#)
- [Trans-Tasman Cooperation on Biosecurity Risk Detection Technology – New Zealand](#)
- [Minister's Media Release – Nov 2018](#)
- [ASF Response Package – Dec 2019](#)

RTT

Source: <https://www.agriculture.gov.au/biosecurity-trade/policy/australia/biosecurity-3d-x-ray>